AthenaK CGL–Landau-Fluid STS Validation Note

April 24, 2026

1 Purpose

This note documents the AthenaK CGL–Landau-fluid (CGL–LF) heat-flux update implemented on branch `CGL-STS-LF`. The goal is not to reproduce the full nonlinear turbulence campaign of Squire et al. (2023), but to show that the implemented method follows the same finite-volume variable strategy and passes method-level checks, including a one-dimensional oblique linear initial-value problem using the Figure 17 background state.

2 Conserved Variables

The hyperbolic CGL system evolves total energy and the conserved anisotropy variable

$$A = \rho \log\left[\frac{p_\perp}{p_\parallel}\frac{\rho^2}{|B|^3}\right]. \tag{1}$$

The legacy array slot is still named `IMU`, with `IAN` used as an alias in CGL code. The Riemann solve remains in A, which avoids the small-$|B|$ pathologies associated with evolving magnetic moment directly.

The LF STS operator temporarily changes only the interpretation of the `IAN` slot:

$$A \to \mu_B = \frac{p_\perp}{|B|}. \tag{2}$$

The standard conserved-variable STS update then advances (E, μ_B) with the LF flux divergence. Immediately after the RKL2 stage, the EOS converts $\mu_B \to A$, and the normal CGL conserved-to-primitive recovery restores $(p_\parallel, p_\perp)$.

3 Landau-Fluid Closure

The implemented closure uses the Snyder–Hammett–Dorland 3+1 heat fluxes,

$$q_\parallel = -\chi_\parallel \rho \nabla_\parallel\left(\frac{p_\parallel}{\rho}\right), \tag{3}$$

$$q_\perp = -\chi_\perp\left[\rho\nabla_\parallel\left(\frac{p_\perp}{\rho}\right) - p_\perp\left(1-\frac{p_\perp}{p_\parallel}\right)\frac{\nabla_\parallel|B|}{|B|}\right], \tag{4}$$

with the finite-collision Squire et al. coefficients

$$\chi_\perp = \frac{2c_\parallel^2}{\sqrt{2\pi}\,c_\parallel|k_\parallel| + \nu_{\rm eff}}, \qquad \chi_\parallel = \frac{8c_\parallel^2}{\sqrt{8\pi}\,c_\parallel|k_\parallel| + (3\pi-8)\nu_{\rm eff}}. \tag{5}$$

Here $\nu_{\rm eff}$ is the background collision frequency plus any active mirror/firehose limiter scattering at the face. In the collisionless limit, these reduce to $\chi_\perp = \sqrt{2/\pi}\, c_\| /|k_\||$ and $\chi_\| = \sqrt{8/\pi}\, c_\|/|k_\||$. Here `lf_k_parallel` is the closure wavenumber $|k_\||$, not a conductivity. In local mode, $c_\| = \sqrt{p_\|/\rho}$ is computed from the live face state. In background mode, only this prefactor is frozen to `lf_c_parallel0`; all gradients and thermodynamic variables remain live.

The face fluxes coupled into STS are

$$F_E^i = \hat{b}^i \left(q_\perp + \frac{1}{2} q_\| \right), \qquad F_{\mu_B}^i = \hat{b}^i \frac{q_\perp}{|B|}. \tag{6}$$

4 Heat-Flux Cap

Following Squire et al. (2023), equations 3.1–3.2, the unlimited heat flux q_L is capped as

$$q = \frac{q_L q_{\rm max}}{q_{\rm max} + |q_L|}. \tag{7}$$

The implemented caps are

$$q_{\perp,\rm max} = \sqrt{\frac{2}{\pi}} c_\| p_\perp, \qquad q_{\|,\rm max} = \sqrt{\frac{8}{\pi}} c_\| p_\|. \tag{8}$$

The cap is applied after computing the unlimited flux with the collision-aware coefficient, so collisions and limiter scattering suppress the flux before the smooth free-streaming bound is applied.

5 Validation Workflow

The quantitative validation is generated by running `scripts/run_cgl_lf_validation.sh` from the repository root. The script runs the CGL–LF unit-style inputs, enables machine-readable diagnostic output from the problem generator, writes logs under `docs/figures/logs`, writes CSV diagnostics under `docs/figures/data`, regenerates the figures in this note, and rebuilds this PDF when `pdflatex` is available.

Each test prints the measured value, reference value or error norm, tolerance, and pass/fail status. The plotted points are read from the same diagnostic CSV files produced by the passing runs.

6 Validation Problems and Tolerances

7 Quantitative Results

Input	Purpose	Assertion
`cgl_transform_check`	$A \leftrightarrow \mu_B$ and primitive recovery	pass
`cgl_lf_quant_parallel`	sinusoidal $T_\parallel$ decay	final Fourier amplitude within 5% of analytic decay
`cgl_lf_quant_parallel _collisional`	sinusoidal $T_\parallel$ decay with $\nu_\mathrm{c} > 0$	final Fourier amplitude within 7% of finite-collision analytic decay
`cgl_lf_quant_perp`	sinusoidal $T_\perp$ decay	final Fourier amplitude within 5% of analytic decay
`cgl_lf_quant_perp _collisional`	sinusoidal $T_\perp$ decay with $\nu_\mathrm{c} > 0$	final Fourier amplitude within 7% of finite-collision analytic decay
`cgl_lf_quant_grad_b`	isolated $\nabla_\parallel \lvert B\rvert$ term	RMS relative error below 10%, positive response correlation
`cgl_lf_flux_limiter`	equation 3.2 heat-flux cap	$\lvert q\rvert \leq q_\mathrm{max}$, sign preserved, update within 20%
`cgl_lf_limiter _heat_flux_suppression`	limiter-collision suppression of heat fluxes	active limiter scattering reduces the unlimited LF flux by more than 20x
`cgl_lf_limiter_mirror`	mirror-biased admissibility stress	intermediate STS states finite, positive, and backup bounded
`cgl_lf_limiter_firehose`	firehose-biased admissibility stress	intermediate STS states finite, positive, and backup bounded
`cgl_lf_field_wave`	field-aligned LF wave reference	complex Fourier mode within 25% of offline reference
`cgl_lf_paper_oblique_wave`	Figure 17 background, LF enabled	guard tolerance 40%; observed errors $< 3 \times 10^{-6}$
`cgl_pure_paper_oblique_wave`	Figure 17 background, no LF	guard tolerance 40%; observed errors $< 3 \times 10^{-6}$
`cgl_pure_paper_eigen_*`	exact pure-CGL Alfvén/slow/fast modes	all tracked Fourier components within 7.5%
`cgl_lf_paper_eigen_*`	exact CGL–LF Alfvén/slow/fast modes	all tracked Fourier components within 7.5%

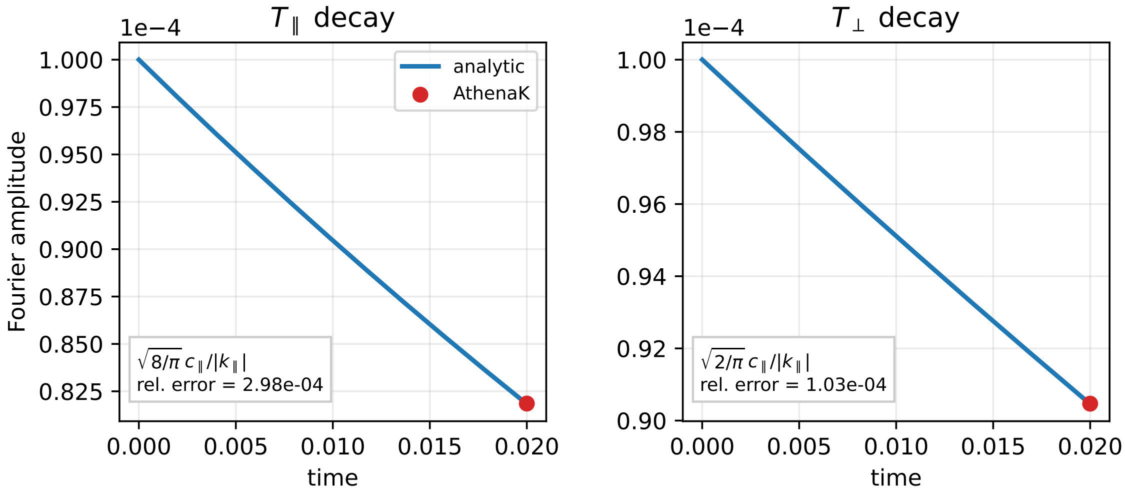

Figure 1: Constant-coefficient decay tests. The $T_\parallel$ case uses $\chi_\parallel = \sqrt{8/\pi}c_\parallel/|k_\parallel|$; the $T_\perp$ case uses $\chi_\perp = \sqrt{2/\pi}c_\parallel/|k_\parallel|$. The red point is the final AthenaK Fourier amplitude and the blue curve is the analytic constant-coefficient decay.

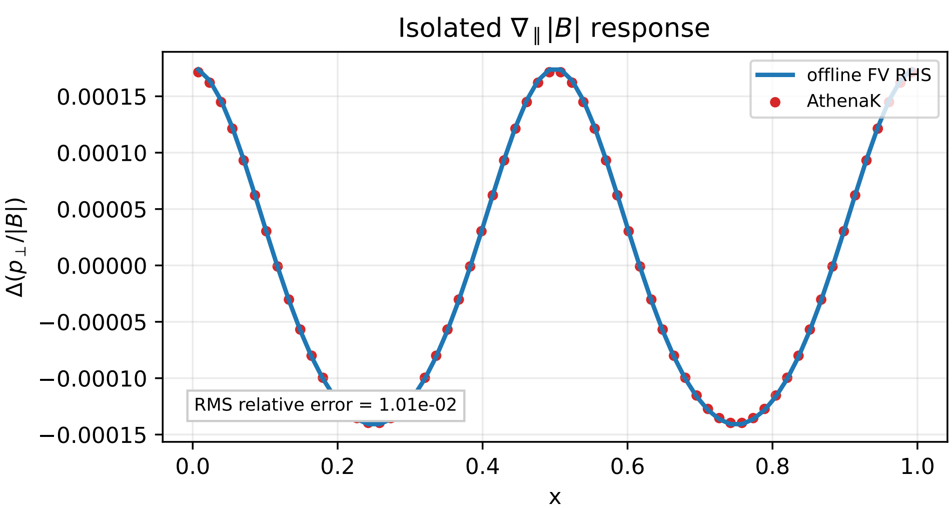

Figure 2: Isolated $\nabla_\parallel|B|$ term in the perpendicular heat flux. The initial thermal gradients are zero; the plotted response is the short-time change in $p_\perp/|B|$ compared with an offline finite-volume reconstruction of the initial LF RHS.

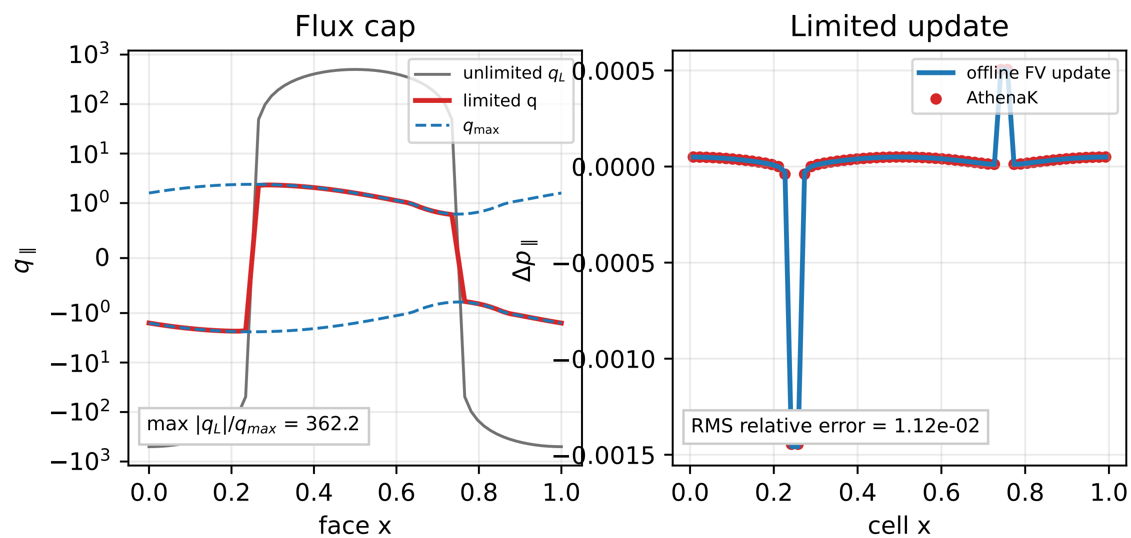

Figure 3: Heat-flux limiter test. The left panel shows the unlimited $q_{\parallel,L}$, the capped $q_\parallel$, and $\pm q_{\parallel,\max}$. The right panel checks that the pressure update produced by the limited flux matches the offline finite-volume update.

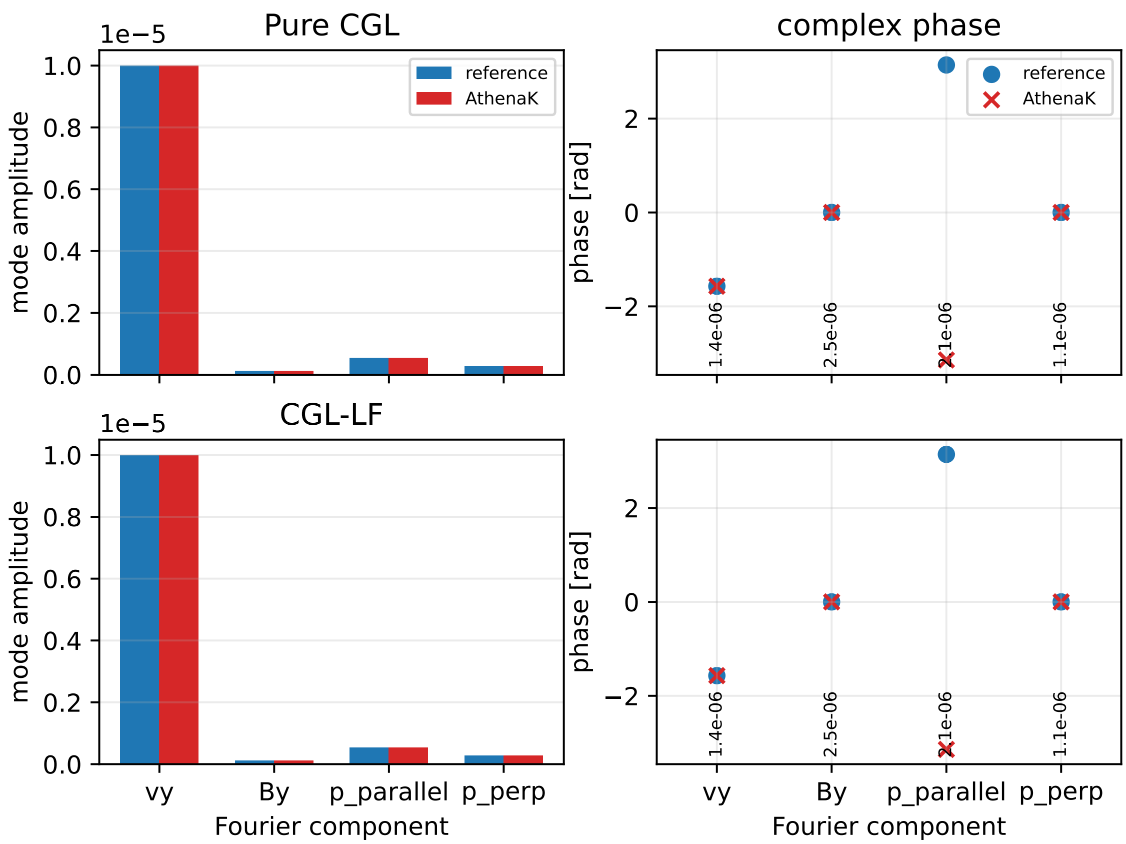

Figure 4: Figure 17-style oblique linear initial-value check. The plotted quantities are the final complex Fourier amplitudes of u_y, B_y, $p_\parallel$, and $p_\perp$ for the pure-CGL and CGL–LF inputs, compared with the offline linearized initial-value reference used by the quantitative pgen. The vertical phase-panel labels are per-variable relative errors.

8 Figure 17-Style Linear Initial-Value Check

The paper-style inputs use the Figure 17 background

$$\rho_0 = 1, \quad p_{\|0} = p_{\perp 0} = 5, \quad u_0 = 0, \quad B_0 = (1, \sqrt{2}, 0.5), \quad k = 2\pi \hat{x}, \quad kL = |k|. \tag{9}$$

The test initializes a small transverse velocity perturbation, not a CGL eigenmode, and compares the final complex Fourier amplitudes of u_y, B_y, $p_\|$, and $p_\perp$ against an offline linearized CGL–LF initial-value reference integrated with the same background state and closure coefficients. Two inputs are provided: one with the LF heat flux enabled and one with pure CGL. This checks the same method class as the Appendix A.2/Figure 17 linear-wave convergence problem.

9 Figure 17-Style Eigenmode Checks

The validation suite now also includes exact eigenmode initializations for the Alfvén, slow-like, and fast-like positive-frequency branches of the same linearized Figure 17 system. The script `scripts/generate_cgl_lf_eigenmode_inputs.py` constructs the linear CGL or CGL–LF matrix for the AthenaK variables $(\rho, u_x, u_y, u_z, B_y, B_z, p_\|, p_\perp)$, diagonalizes it, normalizes the selected eigenvectors, and writes their complex amplitudes and eigenvalues into AthenaK input files. The pgen initializes $\mathrm{Re}[\delta U_k \exp(ikx)]$ and compares the final Fourier mode with $\delta U_k \exp(\lambda t)$, where λ is the supplied eigenvalue. These inputs are the direct replacement for using a single transverse-velocity seed as a proxy for the Squire/Jono linear-wave comparisons.

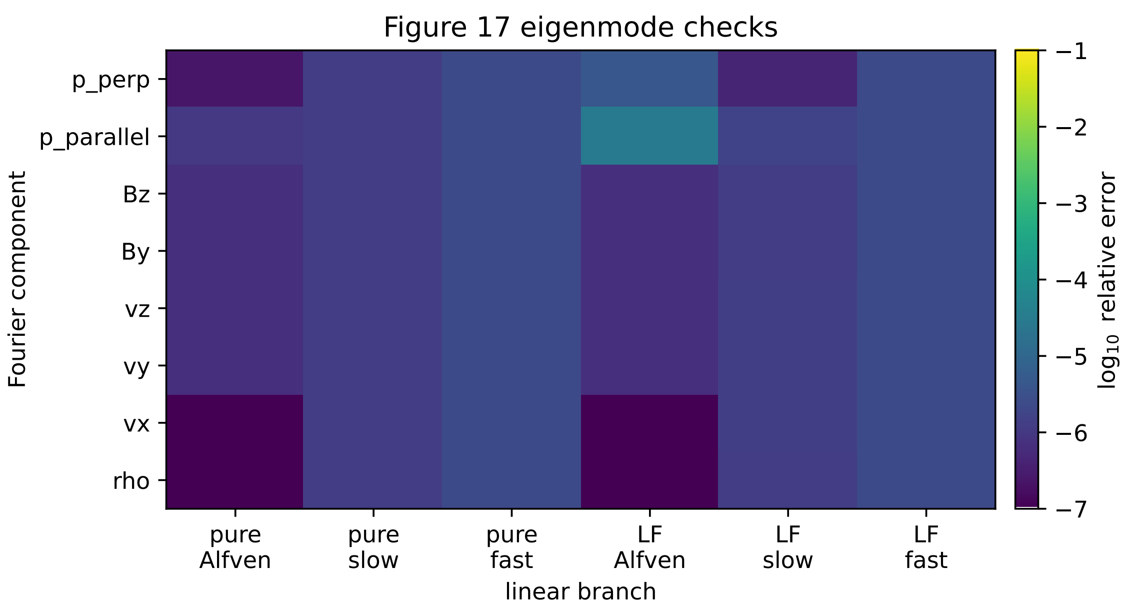

Figure 5: Exact eigenmode checks on the Figure 17 background. Each column is a pure-CGL or CGL–LF Alfvén, slow-like, or fast-like positive-frequency branch. The color scale shows the complex Fourier relative error in each primitive or magnetic component after a short AthenaK evolution compared with the supplied linear eigenmode prediction $\delta U_k \exp(\lambda t)$. The white numbers give the maximum tracked-component error in each branch.

10 Build and Runtime Status

The CPU quantitative build completed successfully. The only build warnings observed were pre-existing variable-length-array warnings in `mesh.cpp`. No CGL-owned files were deleted.

References

[1] J. Squire et al., *Pressure anisotropy and viscous heating in weakly collisional plasma turbulence*, arXiv:2303.00468, 2023.